Forecasting Next-Week Volatility of S&P 500 Stocks

Milestone 2: data wrangling and project redefinition

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CS1090A / AC209A, Fall 2026 Group 38, project 12 TF: Daniel More Torres

The question and why it matters

Ashley

- Returns are close to unpredictable; **volatility clusters**, so it can be forecast.
- Volatility forecasts size positions, set risk limits and price options.
- HAR (Corsi, 2009), a regression on last day, week and month of volatility, is hard to beat.
- **Our question**: with free daily data, does one pooled model with cross-sectional features beat HAR fit stock by stock, out of sample, and in which market regimes?

The data: five public sources (473 MB) and a CRSP check

Source	What	How
fja05680/sp500	S&P 500 members on 2,720 dates since 1996	CSV, pinned
Yahoo Finance	daily OHLCV 2005–2026 for 677 of 950 past and present members	`yfinance`
Alpaca	daily bars from 2016 for 165 members Yahoo lacks; 5-minute bars	REST API
FRED	VIX, 3-month VIX, credit spread, curve slope, T-bill	CSV
SEC EDGAR	earnings-release filing times (8-K Item 2.02), industry codes	JSON API
CRSP (WRDS)	every member's daily prices 2005–2025, for the survivorship check only	SQL, licensed

- Every fetch is cached and re-runnable; no data is committed (licences).
- Code in `src/volforecast`, unit tests incl. no-look-ahead checks, CI on every pull request.

What one row is

Raw table: one ticker on one trading day

3,423,011 rows: `date`, `ticker`, `open`, `high`, `low`, `close`, `adj_close`, `volume`, `source`

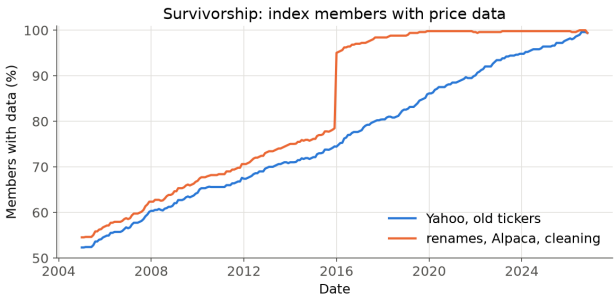
Modeling table: one stock in one week

473,684 rows $\times$ 183 columns, 725 companies, 1,139 weeks

- target: log of next week's Garman–Klass variance (trading hours, with the overnight move in its own column)
- HAR lags: day, week, 4 weeks
- range, overnight, downside variance; returns; liquidity
- market: SPY variance, VIX, rates
- earnings timing, industry; Alpha158 (158 price-volume factors)

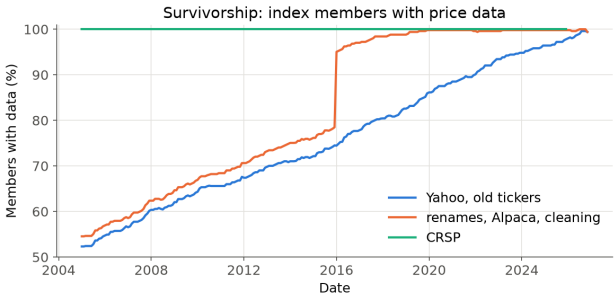
Every feature is known at that Friday's close.

Survivorship: who is missing, and when



- Yahoo drops delisted firms and files renamed ones under the new symbol.
- 30 renames recovered (BK → BNY, ANTM → ELV, …).
- Alpaca fills members that traded in 2016 or later.
- Coverage: 55% in 2005, 99.4% in 2019, 99.8% in 2026.
- Pre-2016 training data leans toward survivors.

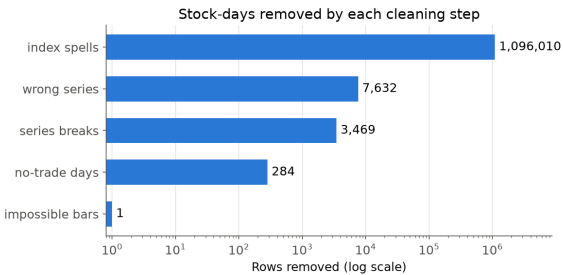
What survivorship costs: a CRSP check



- CRSP (Harvard's WRDS licence), same cleaning: every member, every day, 2005–2025.
- 2005–2015: 191,083 stock-weeks in the free data, 286,597 in CRSP.
- Survivors are calmer: mean log variance 0.04 below CRSP's in the same year before 2016, within 0.01 after.
- From 2019 under 1% of stock-weeks are missing: the test period is barely affected.

Cleaning: each rule, and what it removes

Shuo

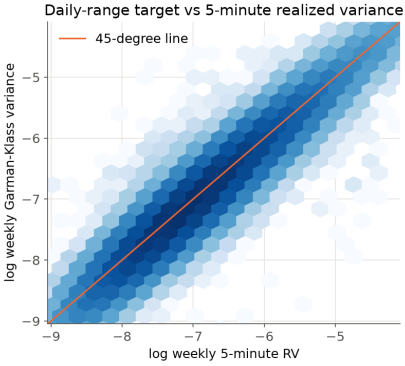


1. **Index spells**: Yahoo reuses tickers (today's `FB` is an ETF).
2. **Wrong series**: 9 hand-checked (Yahoo's PARA trades near $100,000); 4 series that change company (Yahoo's JCI before 2016 is Tyco).
3. **Impossible bars** and **no-trade days**.
4. Prices scaled by `adj_close / close`, as qlib does.

3,423,011 $\rightarrow$ 2,315,615 stock-days. No rule looks at later rows.

Volatility from daily prices lines up with 5-minute data

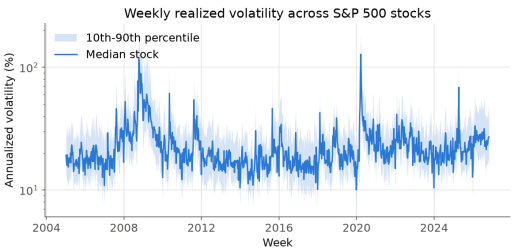
Simon



- **We predict** how much each stock will move next week (its variance).
- **We measure it** from each day's open, high, low and close (Garman–Klass, 1980): $\log \sum_{d \in \text{next week}} \left[\frac{1}{2} \log^2 \frac{H_d}{L_d} - (2 \log 2 - 1) \log^2 \frac{C_d}{O_d} \right]$
- **We checked it** against 5-minute data for 697 companies: correlation 0.95, and 0.94 within a single company.
- **We leave out** the jump from one day's close to the next day's open (the overnight move), a median 27% of a stock's daily variance.

Volatility spikes in a few rare weeks (stress weeks)

Simon



- **Stress week:** VIX, the market's fear gauge, is 30 or more at the week's close, so we know it before we forecast.
- **How often:** 89 of 1,139 weeks, mostly 2008–09, 2020 and 2022.
- **In the test years:** 29 stress weeks, 16 of them in 2020. The tuning years have 1.

How we keep future data out of the forecast (leakage)

Simon

Risk	Control	Enforced by
a feature sees next week's prices	each feature uses only days up to Friday's 4 pm close	`features.py` test: adding later weeks changes no earlier value
earnings news after the 4 pm close	an earnings filing (8-K) after 4 pm counts for the next week	`features.py` test: a 4:30 pm Friday filing counts next week
picking stocks from today's S&P 500 list	each date uses that day's index members	`clean.py` test: a stock counts only between entry and exit
VIX closes at 4:15 pm, after stocks	Friday's 4:15 pm close is used only for next week	`features.py` test: each week gets that Friday's VIX
training weeks run into the test weeks	a one-week gap, as each target looks a week ahead	not yet: the split code comes in Milestone 3

Still in progress: checking when each data source first published its values.

Revised question

Terry

Train 2005–2015, tune 2016–2018, test 2019–2026 with yearly expanding refits.

1. **Accuracy**: do pooled ridge/lasso and gradient-boosted trees beat per-stock HAR on mean QLIKE and MSE? Diebold–Mariano at 5% on the weekly mean loss difference (Newey–West); all models scored on the same rows. A second baseline: HAR pooled across stocks with a constant per stock, to separate the gain from pooling from the gain from features. QLIKE on the exponential of a log forecast gets a training-only smearing correction (Duan, 1983).

2. **Attribution and regimes**: which feature blocks carry any gain (refit without each block), and does it keep its sign in calm and in stress weeks? Holm correction across the family of tests.

Changed: economic-value backtest dropped; earnings from 8-K timestamps; fundamentals deferred (XBRL starts 2009); Alpha158 added; stress defined by VIX, not calendar years.

Staff notes and the AC209A requirement

Terry

Staff note	What we do
omit economic value, Sharpe, drawdown, factor regression	dropped entirely
prioritize regularized regression and boosted trees	the two pooled models of Q1
clustering and neural networks optional	only if time remains after Q1 and Q2
yfinance can fail; delisted stocks missing	cached fetches; renames, wrong series and Alpaca fill; coverage reported
XBRL only from mid-2009	fundamentals deferred
AC209A: one method not covered in class	HAR, GARCH(1,1), QLIKE loss (Patton, 2011), Diebold–Mariano test

Next steps and questions for you

all

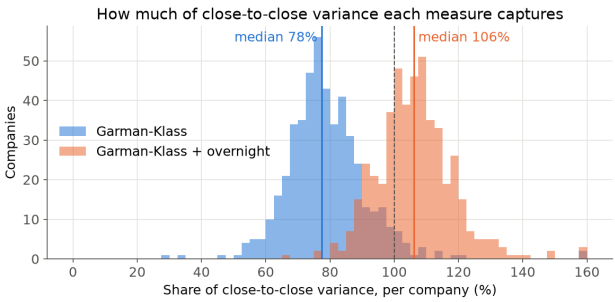
Milestone 3 (proposed owners)

- HAR (per stock, and pooled with a constant per stock) and GARCH baselines: Simon
- ridge and lasso: Shuo
- gradient-boosted trees: Ashley
- QLIKE, MSE, Diebold–Mariano: Terry
- data upkeep and CRSP: Juntang

Questions

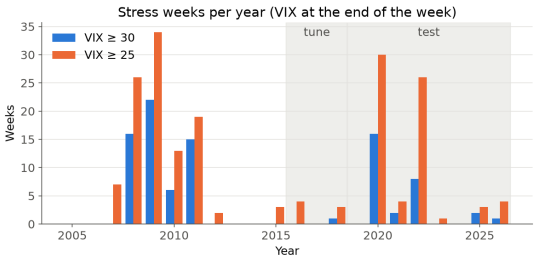
1. May Milestone 3 train on CRSP (WRDS) to 2025 and use free data for 2026?
2. Do HAR, GARCH, QLIKE and Diebold–Mariano meet the AC209A requirement?
3. Yearly expanding refits on 2019–2026, or one fit?
4. Target: trading-hours variance as now, or add the overnight term back?

Backup: what the daily range misses



- 477 companies, 2005–2018 only (train and tune years), days inside the index spell. Each share sums a company's days, so 2008–09 weighs most.
- Overnight is a median 27% of close-to-close variance (19% to 37%, 10th to 90th percentile): 25% in 2005–09, 30% in 2010–18.
- Earnings released after the close show up in this move.
- `rv_overnight` is already a column in the weekly table.

Backup: stress weeks by split



Split	Weeks	VIX $\geq$ 30	episodes	VIX $\geq$ 25
train	573	59	8	104
tune	157	1	1	7
test	409	29	13	68

- Tuning (2016–18) has 1 stress week at VIX $\geq$ 30.
- 16 of the test's 29 stress weeks are in 2020.